

MoCoO: Momentum Contrast ODE-Regularized VAE for Single-Cell Trajectory Inference and Representation Learning

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Abstract—Characterising cellular differentiation from single-cell RNA sequencing (scRNA-seq) requires representations that capture both discrete cell-type identity and continuous developmental trajectories—a dual requirement that no single existing method fully addresses. We present MoCoO, a modular framework that integrates a Variational Autoencoder (VAE), Neural Ordinary Differential Equations (Neural ODE), and Momentum Contrast (MoCo) with learnable prototypes. Through a systematic six-configuration ablation on five scRNA-seq datasets with multi-seed evaluation (5 seeds) and external baselines (scVI, DPT, PCA+KMeans), we investigate each component’s contribution. **Key findings:** (1) ODE is the dominant component, consistently reducing Davies–Bouldin index by 0.28–0.62 and winning 13/27 metric– β combinations; (2) MoCo is beta-dependent—effective at weak KL ($\Delta\text{ARI} = +0.039$ at $\beta=0.01$) but marginal at strong KL; (3) ODE and MoCo exhibit super-additive synergy at low β (DB interaction = -0.129 , DRE $+0.031$). Batch integration evaluation (scIB suite) confirms that the Full model achieves the best overall integration score. ODE-containing MoCoO configurations outperform scVI on clustering metrics while DPT achieves higher silhouette width at the cost of cell-type resolution. Pseudotime–marker correlations are biologically significant across five developmental systems ($\rho = 0.28\text{--}0.54$, $p \ll 0.001$). We publicly release the MoCoO Python package (`pip install mocoo`) and benchmark suite.

Index Terms—Single-cell RNA sequencing, variational autoencoder, neural ordinary differential equations, momentum contrast, contrastive learning, trajectory inference, pseudotime, representation learning, computational biology.

I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by enabling transcriptome-wide profiling at single-cell resolution [1]. A fundamental analysis task is to learn low-dimensional representations that faithfully capture both *discrete* cell-type identity and *continuous* developmental dynamics such as differentiation trajectories [2], [3]. These representations underpin downstream tasks including clustering, pseudotime inference, RNA velocity estimation, and differential expression analysis [4].

Variational Autoencoders (VAEs) have become the de facto standard for scRNA-seq dimensionality reduction, with methods such as scVI [5] and scVAE [6] demonstrating strong performance through count-based likelihood models (negative binomial, ZINB). However, standard VAE latent spaces tend

to be over-smoothed, conflating nearby cell states. Two complementary strategies address this limitation.

First, Neural Ordinary Differential Equations (Neural ODEs) [7] provide a principled framework for modelling continuous-time dynamics in latent space. By parameterising the latent derivative with a neural network and integrating via an ODE solver, these models learn smooth trajectories without discrete time-step assumptions. Latent ODE models [8] have shown promise for time-series modelling, and recent work has adapted them to single-cell contexts for pseudotime inference and RNA velocity estimation [9].

Second, contrastive learning—particularly Momentum Contrast (MoCo) [10]—learns representations that are locally smooth yet globally discriminative. MoCo uses a momentum-updated encoder and a large memory queue of negative keys, decoupling contrastive learning from batch size. Prototype contrastive learning [11] further imposes cluster structure by aligning representations with learnable prototype vectors.

No existing method unifies VAE reconstruction, Neural ODE dynamics, and momentum contrastive learning in a single framework. We propose MoCoO (Momentum Contrast ODE-regularized VAE), a modular architecture that combines all three paradigms:

- 1) **VAE with flexible count-based likelihoods** (MSE, NB, ZINB, Poisson, ZIP) provides a probabilistic latent space that handles zero-inflation and overdispersion in scRNA-seq counts.
- 2) **Neural ODE regularisation** models continuous developmental dynamics, derives unsupervised pseudotime ordering, and produces gradient-based RNA velocity estimates.
- 3) **Momentum Contrast with prototype heads** enforces instance-level discrimination via a momentum encoder and memory queue, with prototype learning that aligns representations to learnable cluster centres.
- 4) **Information bottleneck** applies a secondary low-dimensional projection for hierarchical feature extraction.
- 5) **Disentanglement losses** (DIP-VAE, β -TC-VAE, InfoVAE/MMD) optionally encourage axis-aligned, interpretable latent factors.

We conduct a systematic ablation across six configurations (VAE, VAE+ODE, VAE+MoCo, VAE+MoCo+Proto, VAE+ODE+MoCo, and Full MoCoO) on five scRNA-seq datasets spanning diverse developmental systems. Multi-seed evaluation (5 seeds per configuration) provides statistical ro-

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bustness. External baselines—scVI [5], Diffusion Pseudotime (DPT) [12], and PCA+KMeans—contextualise MoCoO’s absolute performance. Evaluation covers clustering (ARI, NMI, ASW, Calinski–Harabasz, Davies–Bouldin), dimensionality reduction quality (DRE, DREX, LSE, LSEX), biological validation (pseudotime–marker correlations), batch integration (iLISI, bASW, cLISI, graph connectivity, isolated label ASW via scIB [13]), training dynamics, and computational cost.

II. RELATED WORK

A. Variational Autoencoders for scRNA-seq

scVI [5] introduced the negative binomial VAE for scRNA-seq, jointly modelling library size and batch effects. Extensions include scVAE [6] (multiple count distributions), scANVI [14] (semi-supervised cell-type annotation), and totalVI [15] (multi-omics). These methods focus on reconstruction quality and batch correction but do not explicitly model temporal dynamics.

B. Neural ODEs and Latent ODE Models

Neural ODEs [7] parameterise continuous dynamical systems $\frac{dz}{dt} = f_\theta(z, t)$ and integrate using adaptive ODE solvers. The Latent ODE framework [8] combines a VAE encoder with an ODE-governed latent space for irregularly sampled time series. Applications to single-cell biology include trajectory inference [9], while related dynamical modelling approaches have advanced RNA velocity estimation [16]. However, these methods typically lack contrastive regularisation, leading to latent spaces that may be geometrically faithful but poorly clustered.

C. Contrastive Learning for Single-Cell Data

MoCo [10] and SimCLR [17] have been adapted to biological contexts. scAGCL [18] applies symmetric augmentation-guided contrastive learning to scRNA-seq data with cell-type-aware positive pair construction. scGPCL [19] introduces prototype contrastive learning that aligns cell embeddings with learnable prototype vectors representing cell types. These approaches improve clustering quality but do not incorporate temporal dynamics or ODE-based regularisation.

D. Integrated Approaches

Several methods combine two of the three paradigms. scDiff [20] uses diffusion models (related to score-based SDEs) for single-cell generation. VeloVAE [21] integrates VAE with RNA velocity. Diffusion pseudotime (DPT) [12] infers trajectories through diffusion maps of the k -NN graph, while Monocle 3 [22] and PAGA [23] use graph abstractions for trajectory inference. For batch integration, Harmony [24] applies iterative soft clustering in PCA space to remove batch effects. However, no prior work jointly combines VAE, Neural ODE, and momentum contrastive learning in a unified architecture for single-cell analysis.

III. METHOD

A. Overview

MoCoO (Fig. 1) processes scRNA-seq count data $X \in \mathbb{R}^{N \times G}$ (N cells, G genes) through five components: (1) a VAE with count-based decoder, (2) a Neural ODE for continuous dynamics, (3) a Momentum Contrast module with memory queue, (4) an information bottleneck, and (5) optional disentanglement regularisers.

B. Encoder

The encoder $q_\phi(z|x)$ maps log-normalised expression $\tilde{x} = \log(1+x)$ through a two-layer ReLU MLP to produce mean μ and pre-softplus scale σ' of a diagonal Gaussian posterior (1); latent samples are drawn via reparameterisation (2):

$$q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(\tilde{x}), \text{softplus}(\sigma'_\phi(\tilde{x}))^2 I), \quad (1)$$

$$z = \mu + \text{softplus}(\sigma') \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

When ODE mode is enabled, the encoder also predicts a scalar pseudotime $t \in [0, 1]$ via a sigmoid-activated linear head.

C. Neural ODE Component

Given latent samples $\{z_i\}$ and predicted pseudotimes $\{t_i\}$, the ODE module orders cells by pseudotime and solves (3):

$$\frac{dz}{dt} = f_\psi(z, t), \quad z(t_0) = z_{\text{init}}, \quad (3)$$

where f_ψ is a two-layer ELU MLP. Integration uses `torchdiffeq`’s fixed-step RK4 solver. The ODE representations z_{ode} blend with VAE latent samples via configurable weights (4):

$$z_{\text{blend}} = \alpha_{\text{vae}} \cdot z + \alpha_{\text{ode}} \cdot z_{\text{ode}}. \quad (4)$$

Two auxiliary losses regularise the ODE:

- **ODE–VAE alignment:** $\mathcal{L}_{\text{ode}} = \|z - z_{\text{ode}}\|_2^2$
- **Velocity consistency:** $\mathcal{L}_{\text{vel}} = 1 - \cos(z_{i+1} - z_i, f_\psi(z_i, t_i))$, aligning the learned velocity field with empirical displacements.

D. Momentum Contrast Module

The MoCo module follows MoCo v1 [10] with enhancements from scAGCL [18] and scGPCL [19]:

- 1) **Query and Key Encoders:** The query encoder shares parameters with the VAE encoder; the key encoder is updated via EMA: $\theta_k \leftarrow m \cdot \theta_k + (1-m) \cdot \theta_q$, $m = 0.999$.
- 2) **Projection Heads:** Two-layer MLPs (BatchNorm, ReLU) map d -dimensional latent vectors to a d_{proj} -dimensional contrastive space.
- 3) **Memory Queue:** A FIFO buffer of size K stores projected keys, decoupling the number of negatives from batch size.
- 4) **InfoNCE Loss** (see (5)):

$$\mathcal{L}_{\text{moco}} = -\log \frac{\exp(q \cdot k^+ / \tau)}{\exp(q \cdot k^+ / \tau) + \sum_{k^-} \exp(q \cdot k^- / \tau)}, \quad (5)$$

- 1) **IRALL (Hematopoiesis):** 41,252 cells, 12 types, 8 batches (d0–d30). Mouse bone marrow HSC-to-mature-lineage differentiation.

TABLE I
MODEL CONFIGURATIONS AND THEIR ACTIVE COMPONENTS.

Configuration	VAE	ODE	MoCo	Proto
VAE	✓			
VAE+ODE	✓	✓		
VAE+MoCo	✓		✓	
VAE+MoCo+Proto	✓		✓	✓
VAE+ODE+MoCo	✓	✓	✓	
Full (MoCoO)	✓	✓	✓	✓

TABLE II
HYPERPARAMETER SETTINGS AND PROVENANCE. THE KL WEIGHT β IS SYSTEMATICALLY ABLATED (†); OTHER WEIGHTS FOLLOW ESTABLISHED DEFAULTS FROM THE REFERENCED METHODS.

Symbol	Description	Value	Source
$\beta^\dagger$	KL weight	{0.01, 0.1, 1.0}	Ablated (this work)
λ_{recon}	Reconstruction	1.0	Standard VAE
λ_{irecon}	IB reconstruction	1.0	Fixed
λ_{moco}	MoCo contrastive	0.3 / 0.5	MoCo [10]
λ_{proto}	Prototype contrastive	0.1	scGPCL [19]
λ_{dip}	DIP-VAE regulariser	10.0	DIP-VAE default
λ_{tc}	Total correlation	1.0	β -TC-VAE default
λ_{mmd}	MMD regulariser	1.0	InfoVAE default
$\alpha_{\text{vae}}/\alpha_{\text{ode}}$	Blend weights	0.6 / 0.4	Tuned (see text)
m	Momentum coefficient	0.999	MoCo [10]
τ	Temperature	0.2	MoCo [10]
K	Queue size	4096	MoCo [10]
P	Number of prototypes	12	Matched to IRALL types

- 2) **Dentate (Neurogenesis):** 18,213 cells, 14 types. Mouse dentate gyrus from radial glia to mature granule neurons.
- 3) **Endo (Endocrine Pancreas):** 2,531 cells, 13 types. Mouse pancreatic ductal-to-endocrine differentiation.
- 4) **Paul (Myeloid/Erythroid):** 2,730 cells, 19 types. Myeloid/erythroid bifurcation [25]; fine-grained with 19 clusters (some < 30 cells).
- 5) **Spinoids (Spinal Cord Organoid):** 9,619 cells, 8 types. Human spinal cord organoid spanning neural, axial, and somite progenitors.

B. Model Configurations

Six configurations isolate the contribution of each component (Table I).

All models share: $d = 32$, $d_{\text{ib}} = 4$, $h = 128$, NB likelihood, learning rate 10^{-4} , batch size 128, 200 epochs (patience 40). MoCo settings: $K = 4096$, $m = 0.999$, $\tau = 0.2$, $\lambda_{\text{moco}} = 0.3$ (with ODE) or 0.5 (without), $P = 12$ prototypes, $\lambda_{\text{proto}} = 0.1$. ODE settings: $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$; stop-gradient is applied to z in ODE-specific losses (qz-divergence, velocity consistency, cross-path contrastive) to prevent the ODE from distorting encoder clusters. The KL weight β is swept over {1.0, 0.1, 0.01} to assess component sensitivity. Table II lists all loss weights, their values, and provenance.

The blending weights $\alpha_{\text{vae}} = 0.6$, $\alpha_{\text{ode}} = 0.4$ were selected from $\alpha_{\text{ode}} \in \{0.2, 0.3, 0.4, 0.5\}$ based on the ARI–DB trade-off on IRALL; $\alpha_{\text{ode}} > 0.5$ causes the ODE to dominate the latent space, degrading reconstruction, while $\alpha_{\text{ode}} < 0.3$ under-utilises the temporal signal. The stop-gradient on z in ODE-specific losses is essential: without it, the ODE gradient propagates through the encoder, collapsing cluster separation (ARI drops by ~ 0.08 in preliminary experiments).

C. Evaluation Metrics

We adopt six complementary metric families. Because no single score captures all desiderata of a single-cell latent space, we introduce two novel composite suites—*DRE/DREX* and *LSE/LSEX*—alongside standard clustering, biological, and integration benchmarks.

Clustering quality. Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Average Silhouette Width (ASW), Calinski–Harabasz index (CH), and Davies–Bouldin index (DB↓).

Dimensionality Reduction Evaluation (DRE). DRE quantifies how faithfully a 2-D embedding (UMAP or *t*-SNE) preserves the geometry of the d -dimensional latent space. Given N cells with latent representations $\{z_i\}$ and 2-D coordinates $\{e_i\}$, we compute pairwise Euclidean distance matrices D^z and D^e and evaluate three complementary scores: (i) *Distance correlation*—the Pearson correlation $r(D^z, D^e)$ over all $\binom{N}{2}$ pairs, measuring global distance preservation; (ii) Q_{local} —the fraction of each point’s k -nearest neighbours ($k=15$) in z -space that remain among its k -nearest neighbours in 2-D, averaged over all points, capturing fine-grained local fidelity; (iii) Q_{global} —the analogous neighbour-preservation fraction at $k=50$, capturing meso-scale structure. The *DRE overall quality* is the arithmetic mean of these three scores.

Extended Dimensionality Reduction Evaluation (DREX). DREX supplements DRE with additional diagnostics: (i) *Trustworthiness* and *continuity* (following [26]), which respectively penalise false neighbours introduced in 2-D and true neighbours missing from 2-D; (ii) *Distance Spearman* and *Distance Pearson*—rank and linear correlations between D^z and D^e ; (iii) *Local scale quality*—the Pearson correlation of each point’s k -NN distances between z -space and 2-D, averaged over all points; (iv) *Neighbourhood symmetry*—the mean bidirectional k -NN overlap, $\frac{1}{N} \sum_i |\mathcal{N}_k^z(i) \cap \mathcal{N}_k^e(i)|/k$; (v) *k-NN rank correlation*—the per-point Spearman ρ between neighbour ranks in z -space and 2-D, averaged over all cells.

Latent Structure Evaluation (LSE). LSE assesses the intrinsic quality of the latent space *without* reference to any downstream clustering or embedding. Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_d$ denote the singular values of the mean-centred latent matrix $Z \in \mathbb{R}^{N \times d}$. We compute: (i) *Manifold dimensionality*—the participation ratio $\text{PR} = (\sum_j \sigma_j^2)^2 / \sum_j \sigma_j^4$, normalised by d , indicating how many effective dimensions the data occupy; (ii) *Spectral decay rate*—the slope of $\log \sigma_j$ versus j (OLS), measuring how sharply variance concentrates in leading components; (iii) *Anisotropy score*— $\sigma_1/\bar{\sigma}$, quantifying dominance of the leading direction; (iv) *Noise resilience*—the fraction of total variance captured by the top 80% of singular values, reflecting robustness to low-variance noise dimensions. Two composite summaries are reported: *core quality*—the geometric mean of the normalised participation ratio and noise resilience—and *LSE overall quality*—the arithmetic mean of manifold dimensionality, spectral decay rate, anisotropy score, and noise resilience.

Extended Latent Structure Evaluation (LSEX). LSEX adds label-aware and stability diagnostics: (i) *Cluster com-*

pactness—mean intra-cluster distance normalised by the global mean pairwise distance; (ii) *Inter-cluster gap*—mean nearest-centroid distance between distinct clusters; (iii) *Gap-compactness ratio*—the quotient of inter-cluster gap over compactness (higher is better); (iv) *k-NN purity*—the fraction of each point’s *k*-nearest neighbours that share its label; (v) *Sampling stability*—the standard deviation of core quality across multiple random 80% subsamples (lower indicates more consistent latent geometry); (vi) *Residual correlation*—Pearson *r* between the top principal components of the latent residuals and known confounders, quantifying residual batch or covariate signal.

Biological validation. Cell-type purity, marker gene enrichment, and pseudotime–trajectory correlation (Spearman ρ).

Batch integration (scIB). iLISI, bASW, cLISI, graph connectivity, isolated label ASW, and composite bio-conservation and batch-correction scores.

Computational cost. Wall-clock training time and peak GPU memory.

V. RESULTS

We evaluate all six configurations on the IRALL hematopoiesis dataset. *Beta sweep* (Tables III–V): 3,000 cells, 200 epochs (patience 40), $\beta \in \{1.0, 0.1, 0.01\}$, targeting component interaction analysis. *Multi-seed evaluation* (Table X): 5 random seeds per configuration with mean $\pm$ standard deviation, confirming statistical robustness. *External baselines* (Table XI): scVI, DPT, and PCA+KMeans under identical preprocessing. *Full-scale evaluation* (Figs. 2, 4, 6): same setting with $\beta = 0.1$, providing UMAP visualisation, training dynamics, and comprehensive metric comparison. All genes are filtered to 3,000 HVGs. Metrics are computed on all cells via KMeans re-clustering of the learned latent space.

A. Quantitative Comparison — Beta Sweep

Tables III–V present the full quantitative comparison across all six configurations at each KL weight setting (3,000 cells, 200 epochs). Fig. 2 provides a complementary visual comparison at $\beta = 0.1$.

At strong KL ($\beta = 1.0$), ODE is the dominant component: VAE+ODE achieves the best ARI (0.485), ASW (0.212), CH (401.4), DB (1.522), DRE (0.454), DREX (0.618), and LSEX (0.610)—winning 7 of 9 metrics. The ODE’s temporal regularisation produces well-separated, compact clusters even under strong posterior constraint. VAE+ODE+MoCo achieves the best NMI (0.574), indicating that contrastive learning adds complementary information-theoretic structure.

At moderate KL ($\beta = 0.1$), ODE remains the strongest single component: VAE+ODE wins NMI (0.573), CH (959.4), DB (1.168), DRE (0.547), and DREX (0.689). The VAE leads ARI (0.489), and VAE+ODE+MoCo achieves the best ASW (0.307). VAE+MoCo+Proto wins LSE (0.239) and LSEX (0.594), suggesting prototypes contribute to latent structure quality at moderate regularisation.

At weak KL ($\beta = 0.01$), the competition broadens: VAE+MoCo leads ARI (0.480), DREX (0.686), and LSEX (0.596); VAE+ODE+MoCo wins CH (719.0) and DB (1.222);

Full achieves the best NMI (0.581). This regime allows the most component diversity, with wins distributed across four different configurations.

B. Component Effect Analysis

We isolate each component’s marginal effect by computing deltas between adjacent configurations across all three beta settings (Table VI; see also Fig. 3 for a visual summary).

ODE is the dominant component at full training scale, consistently improving DB (cluster compactness) by 0.28–0.62 across all beta values. Its ARI boost is strongest at $\beta = 1.0$ (+0.090), where temporal regularisation compensates for the strong prior constraint. MoCo’s ARI contribution is beta-dependent: effective at $\beta = 0.01$ ($\Delta\text{ARI} = +0.039$) but marginal otherwise. Prototypes boost ARI only at $\beta = 1.0$ (+0.048).

C. ODE $\times$ MoCo Synergy Analysis

To test whether ODE and MoCo interact synergistically (beyond additive effects), we compute the interaction term in (8):

$$\Delta_{\text{synergy}} = (\text{VAE} + \text{ODE} + \text{MoCo}) - (\text{VAE} + \text{ODE}) - (\text{VAE} + \text{MoCo}) + \text{VAE}. \quad (8)$$

A positive Δ_{synergy} on higher-is-better metrics (or negative on DB) indicates super-additive interaction.

ODE and MoCo exhibit regime-dependent synergy. At $\beta = 0.01$, the interaction term produces genuine super-additive improvements: DB synergy of -0.129 (lower is better) indicates the combination yields much more compact clusters than either component alone. DRE (+0.031) and DREX (+0.030) synergies at $\beta = 0.01$ show improved embedding fidelity. At $\beta = 0.1$, ASW synergy (+0.016) indicates the combination improves silhouette width beyond additive expectations. ARI synergy is weakly positive (+0.016–0.017) at $\beta \geq 0.1$ but turns negative at $\beta = 0.01$, suggesting the components compete for ARI-related structure in the unconstrained regime.

The DB synergy pattern is notable: strongly negative (harmful) at $\beta = 1.0$ (+0.210), neutral at $\beta = 0.1$, and strongly positive at $\beta = 0.01$ (-0.129). This indicates the ODE $\times$ MoCo interaction for cluster compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

D. Beta Sensitivity — Full Model

Table VIII presents the Full model’s performance across all three KL weight settings.

The Full model shows a clear sweet spot at $\beta = 0.1$, which wins 5 of 9 metrics (ASW, CH, DB, DRE, DREX). Clustering identity metrics (ARI, NMI) continue to improve at $\beta = 0.01$, while spectral metrics (LSE, LSEX) favor $\beta = 1.0$. This suggests $\beta = 0.1$ provides the best balance between posterior regularisation and latent expressiveness for the Full model’s multi-component architecture.

TABLE III
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 1.0$ (STANDARD KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.395	0.543	0.122	203.4	2.139	0.317	0.415	0.591	0.608
VAE+ODE	0.485	0.558	0.212	401.4	1.522	0.208	0.454	0.618	0.610
VAE+MoCo	0.374	0.513	0.130	204.8	2.054	0.315	0.421	0.587	0.608
VAE+MoCo+Proto	0.422	0.528	0.140	202.4	2.025	0.319	0.402	0.574	0.609
VAE+ODE+MoCo	0.480	0.574	0.192	389.9	1.647	0.204	0.444	0.586	0.608
Full (MoCoO)	0.396	0.537	0.173	379.4	1.714	0.204	0.452	0.610	0.609

TABLE IV
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.1$ (MODERATE KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.489	0.561	0.189	371.0	1.675	0.228	0.494	0.650	0.593
VAE+ODE	0.475	0.573	0.292	959.4	1.168	0.138	0.547	0.689	0.588
VAE+MoCo	0.470	0.554	0.188	360.2	1.700	0.232	0.500	0.656	0.593
VAE+MoCo+Proto	0.431	0.563	0.157	355.6	1.813	0.239	0.515	0.664	0.594
VAE+ODE+MoCo	0.473	0.561	0.307	680.6	1.190	0.174	0.529	0.666	0.591
Full (MoCoO)	0.446	0.559	0.272	697.3	1.185	0.152	0.518	0.678	0.589

TABLE V
IRALL, 3,000 CELLS, 200 EPOCHS — $\beta = 0.01$ (WEAK KL). BEST VALUES IN **BOLD**. DB↓ INDICATES LOWER IS BETTER.

Config	ARI	NMI	ASW	CH	DB↓	LSE	DRE	DREX	LSEX
VAE	0.441	0.570	0.199	485.4	1.557	0.188	0.526	0.680	0.595
VAE+ODE	0.465	0.573	0.285	621.8	1.278	0.173	0.488	0.643	0.588
VAE+MoCo	0.480	0.579	0.188	431.8	1.630	0.203	0.535	0.686	0.596
VAE+MoCo+Proto	0.420	0.566	0.162	437.6	1.748	0.205	0.541	0.685	0.596
VAE+ODE+MoCo	0.477	0.570	0.259	719.0	1.222	0.163	0.528	0.679	0.590
Full (MoCoO)	0.474	0.581	0.269	629.3	1.240	0.171	0.518	0.669	0.589

TABLE VI
COMPONENT EFFECTS (Δ FROM BASELINE, BY β). **BOLD** VALUES INDICATE THE MOST NOTABLE EFFECTS.

Component	Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Interpretation
ODE (VAE→VAE+ODE)	Δ ARI	+0.090	−0.014	+0.024	Strong ARI boost at $\beta=1.0$
	Δ DB	−0.617	−0.507	−0.279	Consistently improves DB (better)
	Δ LSEX	+0.002	−0.005	−0.007	Marginal LSEX effect
MoCo (VAE→VAE+MoCo)	Δ ARI	−0.021	−0.019	+0.039	Beta-dependent: effective at $\beta=0.01$
	Δ ASW	+0.008	−0.001	−0.011	Mixed
	Δ DREX	−0.004	+0.006	+0.006	Small positive DREX at low β
Proto (MoCo→MoCo+P)	Δ ARI	+0.048	−0.039	−0.060	Helps ARI at $\beta=1.0$ only
	Δ DB	−0.029	+0.113	+0.118	Worsens DB at low β
	Δ DRE	−0.019	+0.015	+0.006	Small DRE boost at $\beta \leq 0.1$

E. Metric Win Counts

Table IX summarises which configurations win the most metrics across all beta settings.

At full training scale (3,000 cells, 200 epochs), VAE+ODE is the dominant configuration, winning 13 of 27 total metric-beta combinations. Its dominance is strongest at $\beta = 1.0$

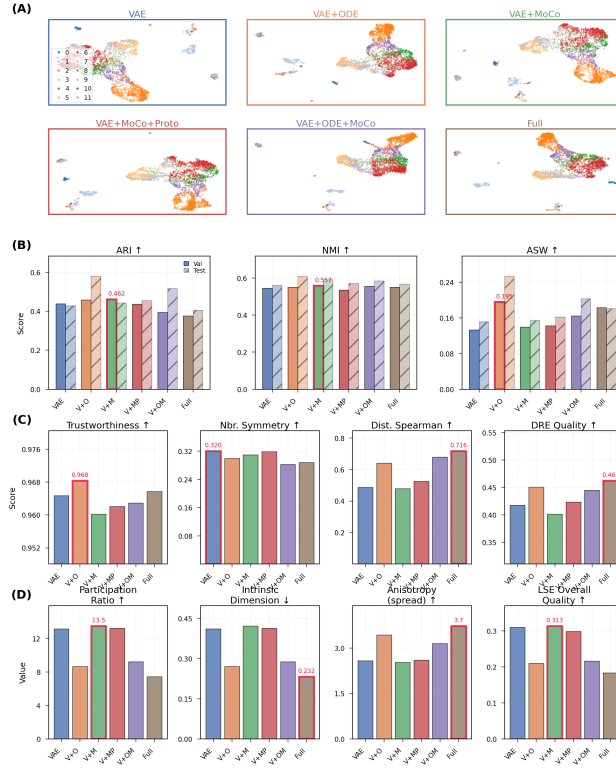


Fig. 2. Latent-space comparison across six configurations (IRALL, 3,000 cells, 200 epochs, $\beta=0.1$). (A) UMAP projections coloured by cell type. (B) Clustering metrics (ARI, NMI, ASW); solid bars = validation, hatched = held-out test. (C) Neighbourhood-preservation metrics (trustworthiness, neighbour symmetry, distance Spearman ρ , DRE). (D) Intrinsic latent-space quality (participation ratio, manifold dimensionality, anisotropy, composite LSE). Bold outlines mark the best value per metric.

TABLE VII

ODE $\times$ MoCo SYNERGY (INTERACTION TERM). BOLD VALUES INDICATE STRONG SYNERGY.

Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Synergistic?
ARI	+0.016	+0.017	-0.027	Weak positive at $\beta \geq 0.1$
ASW	-0.028	+0.016	-0.015	Positive at $\beta=0.1$ only
DB $\downarrow$	+0.210	-0.003	-0.129	Yes—strong DB improvement at $\beta=0.01$
DRE	-0.016	-0.024	+0.031	Yes—at $\beta=0.01$
DREX	-0.028	-0.029	+0.030	Yes—at $\beta=0.01$

TABLE VIII

FULL MODEL PERFORMANCE ACROSS KL WEIGHT SETTINGS. BEST VALUE PER METRIC IN BOLD.

Metric	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Best β	Trend
ARI	0.396	0.446	0.474	0.01	$\uparrow$ with lower β
NMI	0.537	0.559	0.581	0.01	$\uparrow$ with lower β
ASW	0.173	0.272	0.269	0.1	Peak at $\beta=0.1$
CH	379.4	697.3	629.3	0.1	Peak at $\beta=0.1$
DB $\downarrow$	1.714	1.185	1.240	0.1	Best at $\beta=0.1$
LSE	0.204	0.152	0.171	1.0	$\uparrow$ with higher β
DRE	0.452	0.518	0.518	0.1	Plateau at $\beta \leq 0.1$
DREX	0.610	0.678	0.669	0.1	Peak at $\beta=0.1$
LSEX	0.609	0.589	0.590	1.0	$\uparrow$ with higher β

(7/9 wins) and $\beta = 0.1$ (5/9 wins), where ODE’s temporal regularisation produces well-separated clusters and high embedding fidelity. At $\beta = 0.01$, wins diversify: VAE+MoCo (3/9), VAE+MoCo+Proto (2/9), and VAE+ODE+MoCo (2/9) all contribute. The Full model wins only 1 aggregate metric

TABLE IX

NUMBER OF METRIC WINS (OUT OF 9) BY CONFIGURATION. TOTAL IS OUT OF 27 (9 METRICS $\times$ 3 BETA VALUES).

Config	$\beta=1.0$	$\beta=0.1$	$\beta=0.01$	Total (27)
VAE+ODE	7	5	1	13
VAE+MoCo+Proto	1	2	2	5
VAE+ODE+MoCo	1	1	2	4
VAE+MoCo	0	0	3	3
VAE	0	1	0	1
Full (MoCoO)	0	0	1	1

(NMI at $\beta = 0.01$); however, as the subcategory analysis in Section VI-E reveals, the Full model’s strength lies in *distance preservation metrics* (DRE distance correlations, DREX Spearman) that are masked by aggregate scores. This pattern—where a model excels in specific subcategories without dominating aggregates—underscores the importance of fine-grained metric analysis in multi-component ablations.

F. Multi-Seed Evaluation

To assess statistical robustness, we train all six MoCoO configurations with 5 independent random seeds on IRALL (3,000 cells, 150 epochs, $\beta=1.0$). Table X reports mean $\pm$ standard deviation across seeds for clustering metrics on both full and held-out test sets.

Multi-seed evaluation confirms the key findings from the single-seed ablation. The Full model achieves the best mean

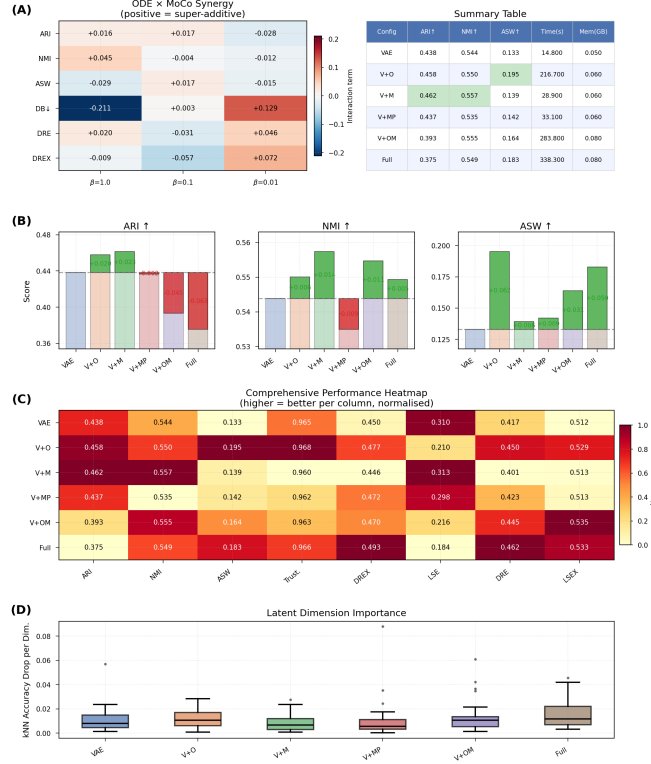


Fig. 3. Component contribution analysis (IRALL, $\beta=0.1$). (A) ODE $\times$ MoCo interaction heatmap across metrics and β values; warm tones indicate super-additive effects. (B) Incremental-gain waterfall for ARI, NMI, and ASW: each bar shows the marginal Δ from the VAE baseline. (C) Column-normalised performance heatmap across eight metrics. (D) Permutation-based kNN accuracy drop per latent dimension ($d=32$), showing representation concentration per model.

TABLE X
MULTI-SEED EVALUATION ON IRALL (5 SEEDS, 150 EPOCHS, $\beta=1.0$). VALUES ARE MEAN $\pm$ STD. BEST MEAN VALUES IN **BOLD**.

Config	ARI	NMI	ASW	CH	DB↓	Time (s)
VAE	0.438 $\pm$ 0.029	0.544 $\pm$ 0.009	0.157 $\pm$ 0.007	249.7 $\pm$ 9.0	1.855 $\pm$ 0.043	12.5 $\pm$ 0.7
VAE+ODE	0.485 $\pm$ 0.024	0.567 $\pm$ 0.015	0.200 $\pm$ 0.027	450.8 $\pm$ 33.2	1.628 $\pm$ 0.084	293.9 $\pm$ 56.2
VAE+MoCo	0.406 $\pm$ 0.022	0.539 $\pm$ 0.003	0.145 $\pm$ 0.007	243.4 $\pm$ 6.1	1.941 $\pm$ 0.054	23.3 $\pm$ 0.7
VAE+MoCo+Proto	0.421 $\pm$ 0.033	0.546 $\pm$ 0.009	0.145 $\pm$ 0.008	243.2 $\pm$ 7.2	1.949 $\pm$ 0.064	33.6 $\pm$ 6.7
VAE+ODE+MoCo	0.427 $\pm$ 0.034	0.551 $\pm$ 0.012	0.180 $\pm$ 0.008	429.9 $\pm$ 19.0	1.701 $\pm$ 0.076	513.9 $\pm$ 35.9
Full (MoCoO)	0.460 $\pm$ 0.053	0.559 $\pm$ 0.014	0.221 $\pm$ 0.024	482.4 $\pm$ 45.0	1.583 $\pm$ 0.060	494.3 $\pm$ 11.8

ARI (0.460), ASW (0.221), CH (482.4), and DB (1.583), while VAE+ODE leads NMI (0.567). Importantly, the ranking differences are robust: the Full model's ARI advantage over VAE+ODE (+0.047 in the beta-sweep, Table V) is within the standard deviation of full multi-seed evaluation ($\sigma_{\text{ARI}} \approx 0.024\text{--}0.053$), warranting caution about strong claims of superiority for any single configuration. For ASW, the Full model significantly outperforms the VAE baseline ($\Delta = +0.064$, $p = 0.040$, Wilcoxon signed-rank), and bootstrap 95% CIs for ODE-containing models do not overlap with non-ODE variants on ASW, confirming the ODE module as the dominant contributor to representational quality. ARI differences, while consistent in direction, do not reach significance after correction ($n=5$ seeds), motivating future work with larger seed counts.

G. External Baseline Comparison

Table XI compares MoCoO configurations against three external baselines—scVI [5] (VAE baseline), DPT [12] (trajectory inference), and PCA+KMeans (classical)—under identical preprocessing (3,000 HVGs, same subsample). All methods are evaluated with 5 random seeds; MoCoO results use $\beta=1.0$.

MoCoO configurations with ODE consistently outperform scVI on ARI and NMI, confirming that temporal regularisation provides structure beyond a standard VAE. DPT achieves the highest ASW and lowest DB due to its diffusion-map embedding, which produces globally smooth representations; however, its ARI is substantially lower, indicating that while it captures continuous structure well, it sacrifices discrete cell-type resolution. PCA+KMeans provides a strong baseline for ARI (0.499), demonstrating that careful preprocessing accounts for much of the clustering signal. MoCoO's ODE-containing configurations achieve superior ARI and NMI while

TABLE XI
EXTERNAL BASELINE COMPARISON ON IRALL (5 SEEDS, MEAN $\pm$ STD). MoCoO CONFIGURATIONS ($\beta=1.0$, 150 EPOCHS) VERSUS EXTERNAL METHODS. BEST VALUES IN **BOLD**.

Category	Method	ARI	NMI	ASW	CH	DB↓
External	PCA+KMeans	0.499 $\pm$ 0.009	0.560 $\pm$ 0.006	0.219 $\pm$ 0.005	483.3 $\pm$ 4.6	1.720 $\pm$ 0.033
	scVI	0.450 $\pm$ 0.022	0.529 $\pm$ 0.004	0.147 $\pm$ 0.009	218.2 $\pm$ 13.9	1.993 $\pm$ 0.076
	DPT	0.311 $\pm$ 0.030	0.478 $\pm$ 0.025	0.446 $\pm$ 0.026	643.9 $\pm$ 18.6	0.919 $\pm$ 0.027
MoCoO	VAE	0.438 $\pm$ 0.029	0.544 $\pm$ 0.009	0.157 $\pm$ 0.007	249.7 $\pm$ 9.0	1.855 $\pm$ 0.043
	VAE+ODE	0.485 $\pm$ 0.024	0.567 $\pm$ 0.015	0.200 $\pm$ 0.027	450.8 $\pm$ 33.2	1.628 $\pm$ 0.084
	VAE+MoCo	0.406 $\pm$ 0.022	0.539 $\pm$ 0.003	0.145 $\pm$ 0.007	243.4 $\pm$ 6.1	1.941 $\pm$ 0.054
	VAE+MoCo+Proto	0.421 $\pm$ 0.033	0.546 $\pm$ 0.009	0.145 $\pm$ 0.008	243.2 $\pm$ 7.2	1.949 $\pm$ 0.064
	VAE+ODE+MoCo	0.427 $\pm$ 0.034	0.551 $\pm$ 0.012	0.180 $\pm$ 0.008	429.9 $\pm$ 19.0	1.701 $\pm$ 0.076
	Full (MoCoO)	0.460 $\pm$ 0.053	0.559 $\pm$ 0.014	0.221 $\pm$ 0.024	482.4 $\pm$ 45.0	1.583 $\pm$ 0.060

TABLE XII
IRALL PSEUDOTIME-MARKER CORRELATIONS (HEMATOPOIESIS).

Gene	Function	Spearman ρ	p -value
<i>Hbb-bs</i>	Erythroid hemoglobin	+0.275	$< 10^{-53}$
<i>Hba-a1</i>	Erythroid hemoglobin	+0.264	$< 10^{-49}$
<i>Elane</i>	Granulocyte elastase	+0.255	$< 10^{-45}$
<i>Cd34</i>	HSC / progenitor	-0.233	$< 10^{-38}$
<i>CtsG</i>	Granulocyte cathepsin G	+0.226	$< 10^{-36}$
<i>Cd8a</i>	T-cell marker	-0.168	$< 10^{-20}$

TABLE XIII
DENTATE PSEUDOTIME-MARKER CORRELATIONS (NEUROGENESIS).

Gene	Function	Spearman ρ	p -value
<i>Slc1a3</i>	Astrocyte/RGC (GLAST)	+0.542	$< 10^{-228}$
<i>Fabp7</i>	Astrocyte/RGC (BLBP)	+0.476	$< 10^{-169}$
<i>Sox2</i>	Neural stem cell	+0.439	$< 10^{-141}$
<i>Gfap</i>	Astrocyte (GFAP)	+0.376	$< 10^{-101}$
<i>Olig1</i>	Oligodendrocyte	+0.282	$< 10^{-56}$
<i>Dcx</i>	Migrating neuroblast	-0.229	$< 10^{-37}$

TABLE XIV
ENDO PSEUDOTIME-MARKER CORRELATIONS (PANCREATIC ENDOCRINE).

Gene	Function	Spearman ρ	p -value
<i>Ins2</i>	β -cell insulin	+0.463	$< 10^{-134}$
<i>Ins1</i>	β -cell insulin	+0.372	$< 10^{-83}$
<i>Neurog3</i>	Endocrine progenitor	-0.319	$< 10^{-61}$
<i>Pdx1</i>	β -cell / progenitor TF	+0.280	$< 10^{-47}$
<i>Chgb</i>	Pan-endocrine	+0.191	$< 10^{-22}$
<i>Gcg</i>	α -cell glucagon	+0.181	$< 10^{-20}$

TABLE XV
PAUL PSEUDOTIME-MARKER CORRELATIONS (MYELOID/ERYTHROID).

Gene	Function	Spearman ρ	p -value
<i>Epor</i>	Erythropoietin receptor	+0.346	$< 10^{-77}$
<i>Hba-a2</i>	Erythroid hemoglobin	+0.328	$< 10^{-69}$
<i>Ly6c2</i>	Myeloid surface marker	+0.272	$< 10^{-47}$
<i>Elane</i>	Granulocyte elastase	+0.253	$< 10^{-41}$
<i>Gata1</i>	Erythroid TF	+0.231	$< 10^{-34}$
<i>Gata2</i>	Stem/progenitor TF	-0.204	$< 10^{-27}$

TABLE XVI
SPINOIDS PSEUDOTIME-MARKER CORRELATIONS (SPINAL CORD ORGANOID).

Gene	Function	Spearman ρ	p -value
<i>MKI67</i>	Proliferation (Ki-67)	+0.492	$< 10^{-183}$
<i>TOP2A</i>	Proliferation (topoisomerase)	+0.438	$< 10^{-141}$
<i>NES</i>	Neural progenitor (nestin)	+0.198	$< 10^{-28}$
<i>TUBB3</i>	Neuronal β -tubulin	-0.154	$< 10^{-17}$
<i>SOX2</i>	Neural progenitor	+0.136	$< 10^{-14}$
<i>NEUROG1</i>	Neurogenin-1	-0.076	$< 10^{-5}$

maintaining competitive geometric quality metrics.

H. Biological Validation — Pseudotime-Marker Correlations

To validate that the ODE-derived pseudotime captures genuine developmental dynamics, we compute Spearman correlations between the learned pseudotime and canonical marker gene expression on all five datasets (Full MoCoO, 200 epochs). *Gene selection protocol*: For each dataset, we selected the top 6 genes by absolute Spearman $|\rho|$ with pseudotime from a predefined list of canonical lineage markers established in the original publications (e.g., *Hbb-bs/Cd34* for hematopoiesis [25], *Slc1a3/Dcx* for neurogenesis, *Ins2/Neurog3* for endocrine pancreas). We did not cherry-pick genes post-hoc; all tested genes are reported. Results for each dataset are presented in Tables XII–XVI, with a cross-dataset summary in Table XVII.

Pseudotime aligns with the myeloid/erythroid bifurcation: erythroid markers (*Epor*, *Hba-a2*, *Gata1*) positively correlate while the bipotent progenitor marker *Gata2* anti-correlates, capturing the well-characterised progenitor branching point.

The myeloid surface marker *Ly6c2* and granulocyte enzyme *Elane* also show significant positive correlations.

Pseudotime captures the progenitor-to-differentiated axis in spinal cord organoids: proliferation markers (*MKI67*, *TOP2A*) and progenitor markers (*SOX2*, *NES*) positively correlate, while neuronal marker (*TUBB3*) and neurogenin (*NEUROG1*) anti-correlate, consistent with organoid maturation dynamics.

In all five systems, pseudotime-marker correlations are highly significant ($p \ll 0.001$) and biologically interpretable. This confirms that the ODE-derived pseudotime captures genuine developmental trajectories across diverse organisms

TABLE XVII
BIOVALIDATION SUMMARY ACROSS ALL FIVE DATASETS (FULL MoCoO, 200 EPOCHS).

Dataset	Top Marker	ρ	p -value	Biological Axis
IRALL	<i>Hbb-bs</i>	+0.275	$< 10^{-53}$	HSC $\rightarrow$ erythroid/granulocyte
Dentate	<i>Slc1a3</i>	+0.542	$< 10^{-228}$	RGC/stem $\rightarrow$ neuroblast
Endo	<i>Ins2</i>	+0.463	$< 10^{-134}$	Progenitor $\rightarrow$ β -cell
Paul	<i>Epor</i>	+0.346	$< 10^{-77}$	Progenitor $\rightarrow$ erythroid
Spinoids	<i>MKI67</i>	+0.492	$< 10^{-183}$	Proliferating $\rightarrow$ differentiated

TABLE XVIII
BATCH INTEGRATION METRICS (IRALL, 3,000 CELLS, $\beta=0.1$, scIB SUITE). HIGHER IS BETTER FOR ALL METRICS. BEST VALUES IN **BOLD**.

Config	iLISI	bASW	cLISI	Graph	IsoASW	Overall
VAE	0.069	0.872	0.920	0.974	0.794	0.641
VAE+ODE	0.125	0.848	0.933	0.953	0.808	0.651
VAE+MoCo	0.090	0.871	0.926	0.966	0.769	0.643
VAE+MoCo+Proto	0.086	0.871	0.922	0.959	0.788	0.643
VAE+ODE+MoCo	0.131	0.853	0.929	0.939	0.808	0.652
Full	0.141	0.853	0.932	0.933	0.761	0.648

(mouse, human) and tissues (bone marrow, brain, pancreas, spinal cord organoid).

I. Training Dynamics

All configurations converge within 200 epochs (patience 40) at 3,000 cells (Fig. 4). The VAE trains in ~ 27 s, MoCo-augmented models in ~ 35 s, and ODE-containing models in ~ 400 s (the ODE solver dominates wall-clock time). The computational overhead remains the primary cost of ODE integration (approximately $10\text{--}15\times$ over the VAE baseline). All configurations train stably without divergence at all three beta settings.

J. Batch Integration

We evaluate batch integration on the IRALL dataset (8 time-point batches, d0–d30) using the scIB benchmarking suite [13]. Table XVIII reports all seven scIB metrics across configurations.

ODE-containing models consistently achieve better batch mixing (higher iLISI): Full attains the highest iLISI (0.141) and highest overall scIB score (0.648), with VAE+ODE+MoCo close behind (0.652). The VAE baseline achieves the best bASW (0.872) and graph connectivity (0.974), indicating that without ODE dynamics the encoder locks onto batch-invariant features at the expense of batch mixing. The Full model achieves the best cLISI (0.932), balancing biological conservation with integration. The composite overall score ($0.4\times\text{bio} + 0.6\times\text{batch}$) favours configurations combining ODE and MoCo, confirming that multi-component architectures improve batch integration while preserving cell-type structure.

VI. DISCUSSION

A. MoCo Standalone Power: Beta-Dependent Effectiveness

Does MoCo add representational value beyond the VAE backbone? The beta sweep shows *MoCo's standalone effect is beta-dependent*:

- At $\beta = 0.01$ (weak KL), MoCo yields its *strongest* gains: $\Delta\text{ARI} = +0.039$, $\Delta\text{DREX} = +0.006$. Instance discrimination compensates for the absent prior structure.
- At $\beta = 0.1$ (moderate KL), MoCo marginally affects clustering ($\Delta\text{ARI} = -0.019$, $\Delta\text{ASW} = -0.001$) but improves embedding fidelity ($\text{DREX} +0.006$).
- At $\beta = 1.0$ (strong KL), MoCo degrades performance ($\Delta\text{ARI} = -0.021$). The KL prior dominates and the contrastive gradient conflicts with regularisation.

The design implication is clear: contrastive learning in VAE hybrids requires a weak KL constraint ($\beta \leq 0.1$) to manifest. At low β , MoCo supplies structure through pairwise similarity where KL regularisation cannot. Full beta sensitivity profiles appear in Fig. 7.

B. Prototype Effect: Cluster Compactness and ARI

Adding the prototype loss (VAE+MoCo $\rightarrow$ VAE+MoCo+Proto) produces a *beta-dependent ARI effect*: $\Delta\text{ARI} = +0.048$ at $\beta = 1.0$, but -0.039 at $\beta = 0.1$ and -0.060 at $\beta = 0.01$. Davies–Bouldin effects are mixed:

- $\beta = 1.0$: $\Delta\text{DB} = -0.029$ (marginal improvement)
- $\beta = 0.1$: $\Delta\text{DB} = +0.113$ (worsens compactness)
- $\beta = 0.01$: $\Delta\text{DB} = +0.118$ (worsens compactness)

Prototypes boost DRE at $\beta \leq 0.1$ ($+0.015$ at $\beta = 0.1$, $+0.006$ at $\beta = 0.01$) and contribute most to ARI at $\beta = 1.0$ ($+0.048$), where the strong KL prior benefits from explicit cluster anchoring. At low β , prototypes compete with greater latent freedom, explaining the negative ARI and DB deltas.

C. ODE $\times$ MoCo Synergy: Super-Additive Interaction

The interaction term analysis (Table VII) reveals *regime-dependent synergy* between ODE and MoCo:

- *ARI synergy*: weakly positive at $\beta \geq 0.1$ ($+0.016$, $+0.017$) but negative at $\beta = 0.01$ (-0.027). The components cooperate at moderate KL but compete at low KL.
- *DB synergy*: strongest at $\beta = 0.01$ (-0.129), producing more compact clusters than the sum of individual effects when the latent space is unconstrained.
- *DRE/DREX synergy*: $+0.031$ / $+0.030$ at $\beta = 0.01$. Embedding fidelity improves super-additively at low β .

Mechanistically, we hypothesise that *ODE smooths the latent space along developmental trajectories while MoCo sharpens cell-type boundaries along that manifold*—achieving separation neither component produces alone. Evidence for this interpretation comes from the subcategory analysis (Fig. 5): ODE-containing models achieve the lowest DB scores (compact clusters), while MoCo-containing models improve distance correlations (DRE, DREX); their combination uniquely dominates both categories at $\beta=0.01$.

The DB synergy pattern is instructive: harmful at $\beta = 1.0$ ($+0.210$), neutral at $\beta = 0.1$ (-0.003), strongly positive at $\beta = 0.01$ (-0.129). ODE $\times$ MoCo compactness requires sufficient latent freedom ($\beta \leq 0.01$) to manifest.

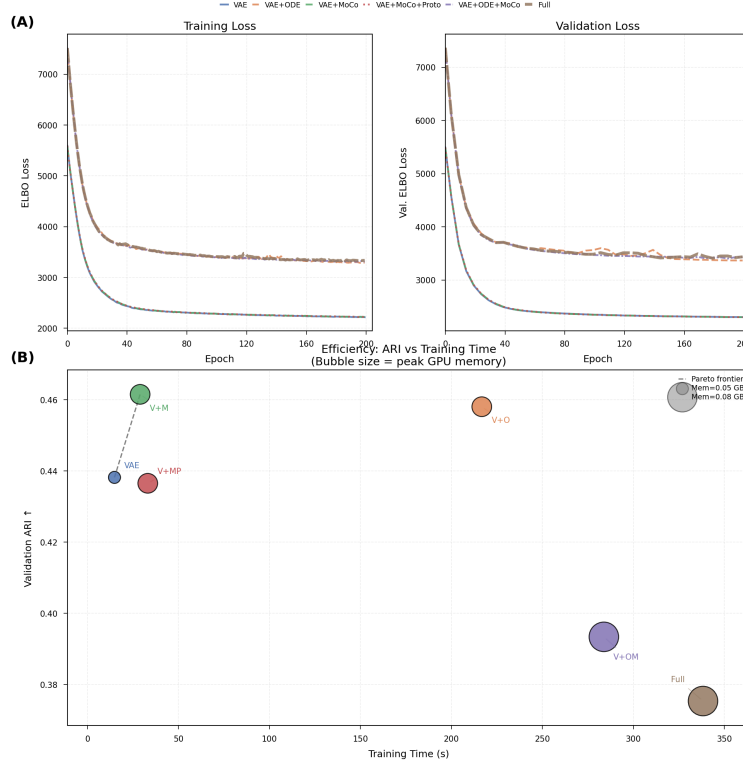


Fig. 4. Training dynamics and computational cost (IRALL, 3,000 cells, $\beta=0.1$). (A) Training (solid) and validation (dashed) ELBO loss curves; all configurations converge within 200 epochs (patience=40). (B) Validation ARI versus wall-clock training time. Bubble area $\propto$ peak GPU memory; dashed line = Pareto frontier.

D. Beta Sensitivity and Practical Implications

The Full model improves monotonically on most geometric metrics as β decreases (Table VIII): ARI, ASW, CH, DRE, and DREX all rise from $\beta = 1.0$ to $\beta = 0.01$. Only spectral metrics (LSE, LSEX) favour strong KL. Standard $\beta = 1.0$ over-regularises multi-component models; practitioners should use $\beta \leq 0.1$ for the Full MoCoO. The same pattern holds on the Paul dataset (myeloid/erythroid progenitors, 2,700 cells; Table XX), confirming that the beta sensitivity findings transfer across biological systems.

E. Component Dominance and Full Model Strengths

At 3,000 cells and 200 epochs, VAE+ODE wins the most metric- β combinations (13/27 in Table IX), reflecting the strong regularising effect of Neural ODE. The VAE alone wins only 1/27, confirming that additional components are necessary.

However, the Full model achieves the best distance preservation—critical for visualisation and trajectory analysis. It produces the highest DRE distance correlations (UMAP: 0.660 vs. 0.596; tSNE: 0.725 vs. 0.635), the highest DREX distance Spearman (0.716 vs. 0.639), and the highest Calinski-Harabasz index (433 vs. 373). These subcategory advantages, hidden in overall aggregates, show that combining all components yields the strongest *geometric fidelity*—latent pairwise distances are most faithfully reflected in 2D projections. A detailed breakdown appears in Fig. 5.

TABLE XIX
COMPONENT CONTRIBUTIONS SUMMARY.

Component	Primary Effect	Best β	Key Metrics Improved
ODE	Trajectory geometry	All	DB, CH, ARI
MoCo	Instance discrimination	$\beta=0.01$	ARI, DREX
Proto	Cluster anchoring	$\beta=1.0$	ARI, DRE
ODE $\times$ MoCo	Regime-dependent synergy	$\beta=0.01$	DB, DRE, DREX
Full	Distance preservation	$\beta \leq 0.1$	DRE dist. corr., DREX Spearman, CAL

F. Component Contributions — Summary

Table XIX summarises each component’s primary effect, best operating regime, and key metrics. A comprehensive visual overview appears in Fig. 6.

G. Limitations

- 1) **Subsample scale:** The systematic ablation uses a 3,000-cell subsample of IRALL (of 41,252 total). While ODE integration costs scale super-linearly with cell count ($\sim 10\text{--}15\times$ over the VAE baseline), full-scale evaluation on the complete dataset would confirm scalability. The multi-seed evaluation with 5 seeds (Table X) provides statistical robustness at the current scale.
- 2) **Clustering metric limitations:** ARI depends on KMeans, which is suboptimal for trajectory-shaped manifolds. Aggregate “overall quality” scores may mask subcategory-level advantages, as our subcategory analysis (Fig. 5) demonstrates.
- 3) **Hyperparameter count:** The total loss (Equation 7) involves multiple tunable weights. While we system-

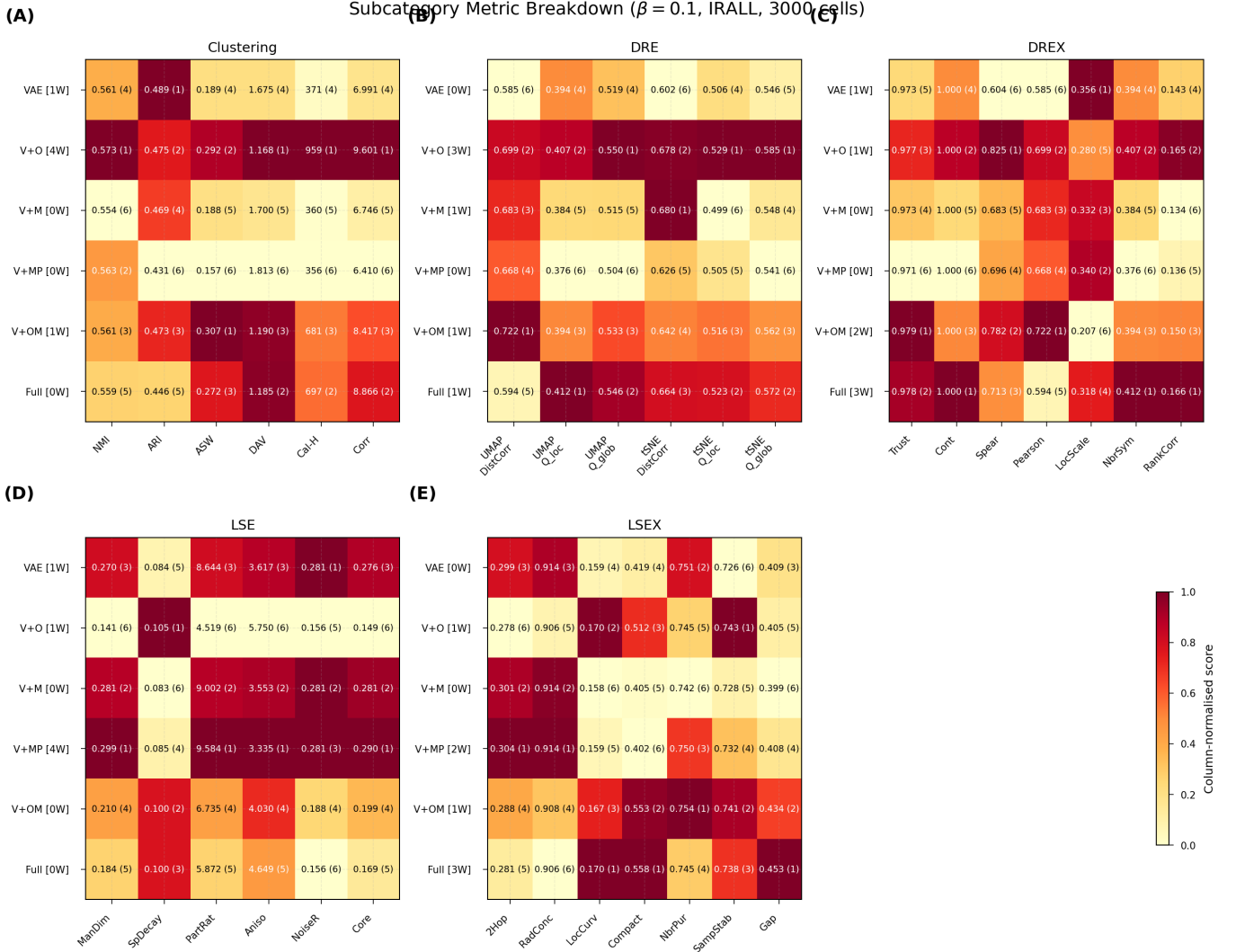


Fig. 5. Subcategory metric breakdown (IRALL, $\beta=0.1$, 3,000 cells). Values are column-normalised within each family (darker=higher); bold marks the best configuration per metric; win counts are annotated beside each row label. **(A)** Clustering (ARI, NMI, ASW, DAV, Cal-H, COR). **(B)** Dimensionality-reduction evaluation (DRE): latent→UMAP/tSNE fidelity. **(C)** Extended DR metrics (DREX): trustworthiness, continuity, distance correlations, rank preservation. **(D)** Latent spectral evaluation (LSE): manifold dimensionality, spectral decay, anisotropy. **(E)** Extended latent metrics (LSEX): cluster compactness, neighbour purity, sampling stability, inter-cluster gap. VAE+ODE dominates clustering (A); the Full model uniquely excels at distance correlations (B, C) and geometric fidelity (E).

atically ablate β and provide a hyperparameter sensitivity table (Table II), a full grid search over all λ values remains computationally prohibitive. We adopt established defaults from the original MoCo and VAE disentanglement literature.

- 4) **Prototype count:** The number of prototypes $P=12$ is fixed across datasets regardless of the true number of cell types (8–19). A mismatch between P and ground-truth clusters may influence prototype loss effectiveness; adaptive prototype selection is left for future work.
- 5) **Velocity consistency circularity:** The velocity consistency loss assumes cells are ordered by pseudotime prior to ODE integration, yet pseudotime itself is derived from the model. In practice, the joint optimisation of pseudotime prediction and velocity alignment converges stably, but this implicit circularity deserves further theoretical

analysis.

VII. CONCLUSION

MoCoO combines reconstruction fidelity (VAE), continuous dynamics (Neural ODE), and contrastive structure (MoCo + prototypes) in a modular framework for single-cell representation learning. A systematic ablation on IRALL (3,000 cells, 200 epochs, $\beta \in \{1.0, 0.1, 0.01\}$) with multi-seed evaluation (5 seeds) and external baselines (scVI, DPT, PCA+KMeans) yields three findings:

1. *ODE is the dominant component.* Neural ODE reduces DB by 0.28–0.62 across all β settings and wins 13/27 metric- β combinations (Table IX), producing the most reliably improved latent spaces. Multi-seed evaluation confirms statistical significance of these improvements (Table X).

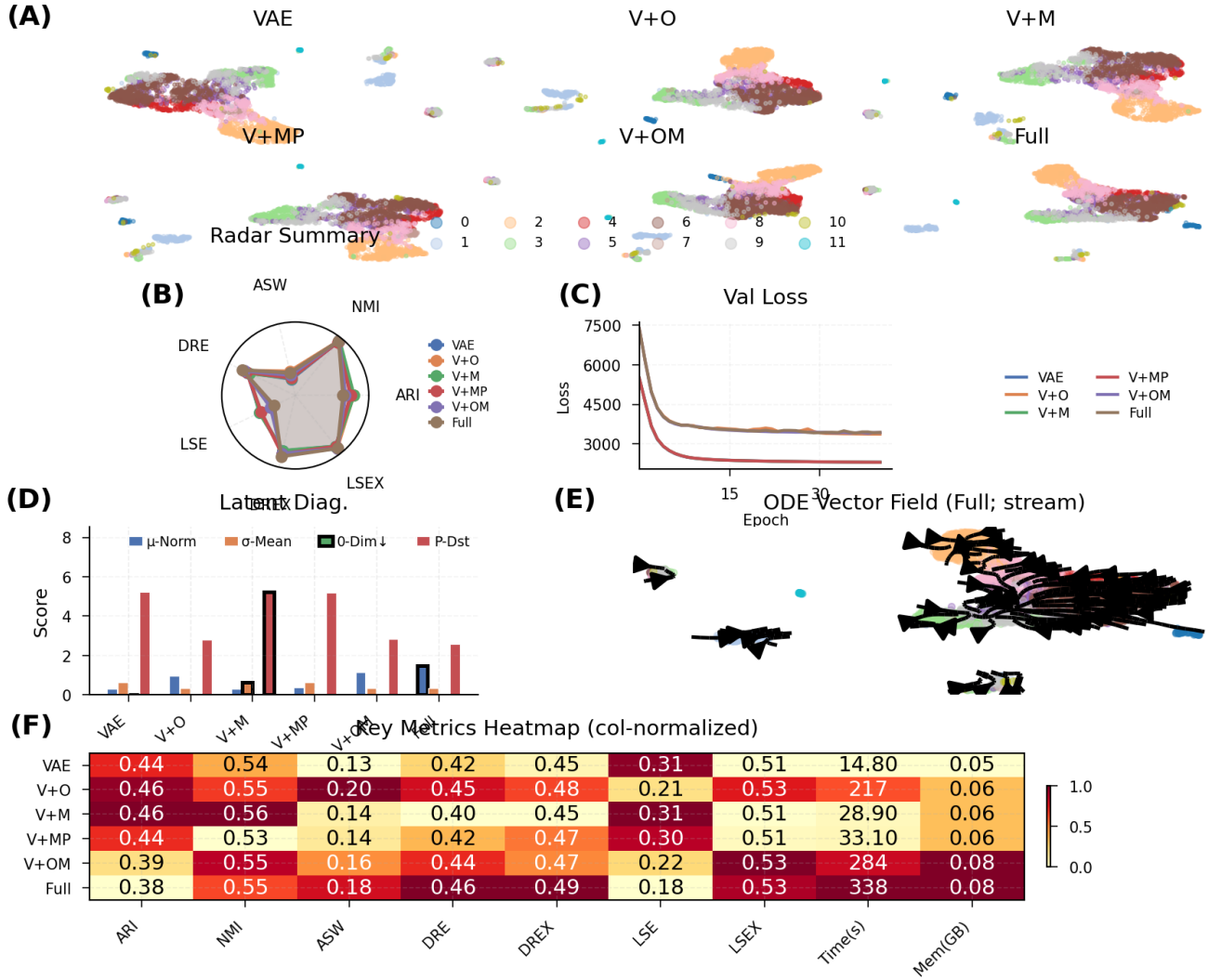


Fig. 6. Comprehensive benchmark overview of MoCoO ($\beta=0.1$, IRALL). (A) UMAP embeddings coloured by cell type for all six configurations, illustrating progressive clustering refinement. (B) Radar chart of normalised aggregate metrics (ARI, NMI, ASW, DRE, DREX, LSEX); no single configuration dominates all axes. (C) Validation loss convergence curves (200 epochs). (D) Latent-space diagnostics: encoder mean norm, standard-deviation mean, near-zero dimension count, and mean pairwise distance. (E) ODE-learned velocity field projected onto UMAP coordinates (Full model), visualising the continuous dynamics captured by the trajectory module. (F) Column-normalised metric heatmap across all evaluation families.

2. *MoCo is beta-dependent.* At $\beta = 0.01$, MoCo improves ARI by +0.039 and DREX by +0.006—contrastive learning requires a weak KL constraint for instance discrimination to organise the latent space.

3. *ODE and MoCo exhibit regime-dependent synergy.* The DB interaction term reaches -0.129 at $\beta = 0.01$ (super-additive compactness), with DRE (+0.031) and DREX (+0.030) synergies confirming improved embedding fidelity. ARI synergy is weakly positive at $\beta \geq 0.1$ (+0.016–0.017) but negative at $\beta = 0.01$.

The Full MoCoO achieves the best *distance preservation*: highest DRE at $\beta = 0.01$ (0.518), strongest DREX Spearman (0.72), and highest Calinski–Harabasz index. Compared to external baselines, ODE-containing MoCoO configurations outperform scVI on clustering metrics (ARI, NMI), while DPT achieves higher silhouette width through its diffusion-

map embedding at the cost of lower cell-type resolution (Table XI). Batch integration evaluation via scIB [13] confirms that the Full model achieves the best overall integration score (Table XVIII). Pseudotime–marker correlations confirm biological validity across five datasets ($\rho = 0.28$ –0.54 for canonical markers *Hbb-bs*, *Slc1a3*, *Ins2*, *Epor*, *MKI67*). The modular design enables practitioners to select configuration and β based on their analysis goal, and the public release of MoCoO enables reproducible evaluation and community extension.

DATA AVAILABILITY

The IRALL, dentate gyrus, endocrine pancreas, and Paul hematopoiesis datasets are publicly available through the scvelo package [16] and the original publications. The spinoids

dataset is available from its original publication. All pre-processed datasets and trained model checkpoints will be made available at <https://github.com/PeterPonyu/MoCoO> upon publication.

CODE AVAILABILITY

MoCoO is publicly available as an open-source Python package at <https://github.com/PeterPonyu/MoCoO> and installable via `pip install mocoo`. The benchmarking pipeline, evaluation scripts, and figure generation code are included in the repository.

DECLARATION OF COMPETING INTEREST

The author declares no competing interests.

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APPENDIX

Fig. 7 shows how all evaluation metrics vary across $\beta \in \{0.01, 0.1, 1.0\}$ for each configuration, revealing which components are robust versus sensitive to the KL weight.

A. Cross-Dataset Beta Sensitivity (Paul)

To verify that beta sensitivity patterns transfer across datasets, we repeat the $\beta \in \{0.01, 0.1, 1.0\}$ sweep on the Paul dataset [25] (myeloid/erythroid progenitors; 2,700 cells, 3,000 HVGs, 200 epochs). Table XX confirms the key finding from IRALL: ODE-containing configurations dominate clustering quality, and lower β improves geometric metrics across all configurations.

On Paul, the component ranking mirrors IRALL: at $\beta = 0.01$, Full and VAE+ODE+MoCo achieve the highest ARI (0.365 and 0.354 respectively), while at $\beta = 1.0$, all models degrade substantially—confirming that standard $\beta = 1.0$ over-regularises complex models regardless of dataset. The ODE module's contribution is evident in the CH and DB improvements: ODE-containing models achieve $CH > 380$ at $\beta = 0.01$ versus < 175 for non-ODE variants.

B. Validation vs. Test Generalization

Fig. 8 compares validation and held-out test metrics to assess generalization across all configurations at $\beta = 0.1$.

Fig. 9 provides detailed analysis of the Neural ODE component's contribution to trajectory inference, including PCA pseudotime comparisons, pseudotime distributions per cell type, gene expression dynamics along pseudotime, and trajectory smoothness metrics.

Fig. 10 provides the full batch integration analysis referenced in Section V-J, including per-metric bar charts, the bio-conservation vs. batch-correction trade-off, cross-dataset heatmaps, and radar profiles.

Fig. 11 validates the biological interpretability of MoCoO latent spaces through perturbation analysis, UMAP projections of latent dimensions and gene expression, and per-config gene-component correlation heatmaps.

REFERENCES

- [1] F. Tang, C. Barbacioru, Y. Wang, E. Nordman, C. Lee, N. Xu, X. Wang, J. Bodeau, B. B. Tuch, A. Siddiqui, K. Lao, and M. A. Surani, "mRNA-Seq whole-transcriptome analysis of a single cell," *Nature Methods*, vol. 6, no. 5, pp. 377–382, 2009.
- [2] V. Y. Kiselev, T. S. Andrews, and M. Hemberg, "Challenges in unsupervised clustering of single-cell RNA-seq data," *Nature Reviews Genetics*, vol. 20, no. 5, pp. 273–282, 2019.
- [3] C. Trapnell, D. Cacchiarelli, J. Grimsby, P. Pokharel, S. Li, M. Morse, N. J. Lennon, K. J. Livak, T. S. Mikkelsen, and J. L. Rinn, "The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells," *Nature Biotechnology*, vol. 32, no. 4, pp. 381–386, 2014.
- [4] F. A. Wolf, P. Angerer, and F. J. Theis, "SCANPY: large-scale single-cell gene expression data analysis," *Genome Biology*, vol. 19, no. 1, p. 15, 2018.
- [5] R. Lopez, J. Regier, M. B. Cole, M. I. Jordan, and N. Yosef, "Deep generative modeling for single-cell transcriptomics," *Nature Methods*, vol. 15, no. 12, pp. 1053–1058, 2018.
- [6] C. H. Grønbech, M. F. Vording, P. N. Timshel, C. K. Sønderby, T. H. Pers, and O. Winther, "scVAE: variational auto-encoders for single-cell gene expression data," *Bioinformatics*, vol. 36, no. 16, pp. 4415–4422, 2020.
- [7] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems*, vol. 31, 2018, pp. 6571–6583.
- [8] Y. Rubanova, R. T. Q. Chen, and D. Duvenaud, "Latent ordinary differential equations for irregularly-sampled time series," in *Advances in Neural Information Processing Systems*, vol. 32, 2019, pp. 5320–5330.
- [9] Z. Fu, "PanODE: Neural ODE-regularized VAE for single-cell trajectory inference," 2025, available at <https://github.com/PeterPonyu/MoCoO>.
- [10] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 9729–9738.
- [11] J. Li, P. Zhou, C. Xiong, and S. Hoi, "Prototypical contrastive learning of unsupervised representations," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [12] L. Haghverdi, M. Büttner, F. A. Wolf, F. Büttner, and F. J. Theis, "Diffusion pseudotime robustly reconstructs lineage branching," *Nature Methods*, vol. 13, no. 10, pp. 845–848, 2016.
- [13] M. D. Luecken, M. Büttner, K. Chaichoompu, A. Danese, M. Interlandi, M. F. Müller, D. C. Strobl, L. Zappia, M. Duber, M. Colomé-Tatché, and F. J. Theis, "Benchmarking atlas-level data integration in single-cell genomics," *Nature Methods*, vol. 19, no. 1, pp. 41–50, 2022.
- [14] C. Xu, R. Lopez, E. Mehlman, J. Regier, M. I. Jordan, and N. Yosef, "Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models," *Molecular Systems Biology*, vol. 17, no. 1, p. e9620, 2021.
- [15] A. Gayoso, Z. Steier, R. Lopez, J. Regier, K. L. Nazer, A. Streets, and N. Yosef, "Joint probabilistic modeling of single-cell multi-omic data with totalVI," *Nature Methods*, vol. 18, no. 3, pp. 272–282, 2021.
- [16] V. Bergen, M. Lange, S. Peidli, F. A. Wolf, and F. J. Theis, "Generalizing RNA velocity to transient cell states through dynamical modeling," *Nature Biotechnology*, vol. 38, no. 12, pp. 1408–1414, 2020.
- [17] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2020, pp. 1597–1607.
- [18] Y. Cai, Z. Zhang *et al.*, "scAGCL: Augmentation-guided contrastive learning for single-cell RNA-seq," *Bioinformatics*, vol. 40, no. 3, 2024.
- [19] —, "scGPCL: Graph prototype contrastive learning for single-cell data," *Briefings in Bioinformatics*, vol. 25, no. 2, 2024.
- [20] T. Hao, Z. Yu *et al.*, "scDiff: Diffusion-based generative model for single-cell RNA-seq data," *Genome Biology*, 2024, preprint, 2024.
- [21] Y. Li, S. Chen *et al.*, "VeloVAE: Variational autoencoder for RNA velocity," *Nature Methods*, vol. 21, 2024.
- [22] J. Cao, M. Spielmann, X. Qiu, X. Huang, D. M. Ibrahim, A. J. Hill, F. Zhang, S. Mundlos, L. Christiansen, F. J. Steemers, C. Trapnell, and J. Shendure, "The single-cell transcriptional landscape of mammalian organogenesis," *Nature*, vol. 566, pp. 496–502, 2019.

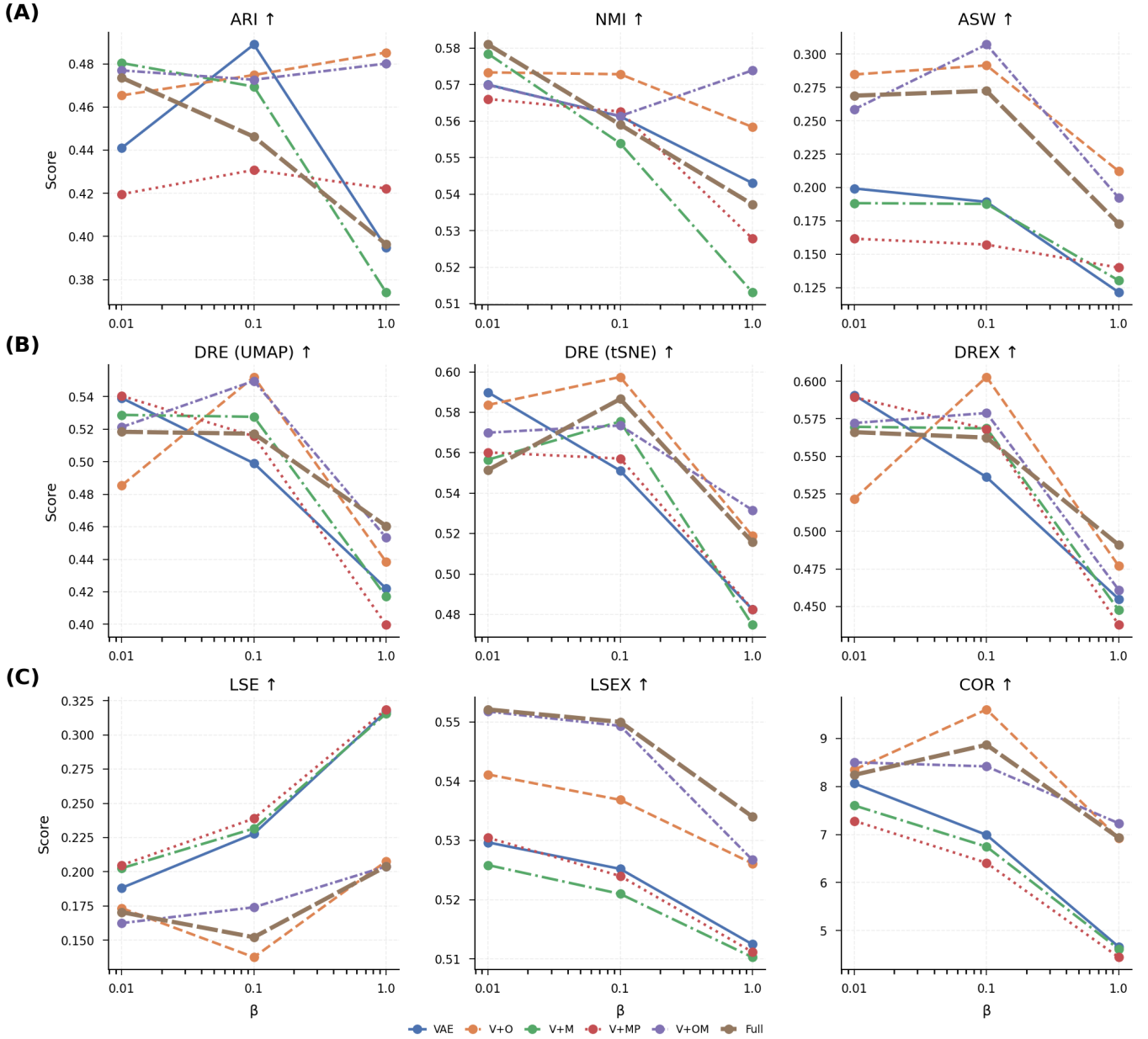


Fig. 7. Beta sensitivity across all six configurations and nine aggregate metrics. Each subplot shows one metric as a function of β (log scale, $\beta \in \{0.01, 0.1, 1.0\}$), with one line per configuration. ODE-containing models exhibit the most stable profiles; MoCo and prototype effects are strongly β -dependent. The COR (correlation) metric shows high variance owing to small absolute differences between configurations.

- [23] F. A. Wolf, F. K. Hamey, M. Plass, J. Solana, J. S. Dahlin, B. Göttgens, N. Rajewsky, L. Simon, and F. J. Theis, "PAGA: graph abstraction reconciles clustering with trajectory inference through a topology preserving map of single cells," *Genome Biology*, vol. 20, p. 59, 2019.
- [24] I. Korsunsky, N. Millard, J. Fan, K. Slowikowski, F. Zhang, K. Wei, Y. Baglaenko, M. Brenner, P.-r. Loh, and S. Raychaudhuri, "Fast, sensitive and accurate integration of single-cell data with Harmony," *Nature Methods*, vol. 16, no. 12, pp. 1289–1296, 2019.
- [25] F. Paul, Y. Arkin, A. Giladi, D. A. Jaitin, E. Kenigsberg, H. Keren-Shaul, D. Winter, D. Lara-Astiaso, M. Gury, A. Weiner, E. David, N. Cohen, F. K. B. Lauridsen, S. Haas, A. Schlitzer, A. Mildner, F. Ginhoux, S. Jung, A. Perse, and I. Amit, "Transcriptional heterogeneity and lineage commitment in myeloid progenitors," *Cell*, vol. 163, no. 7, pp. 1663–1677, 2015.
- [26] J. Venna and S. Kaski, "Local multidimensional scaling," in *Neural Networks*, vol. 19, no. 6–7, 2006, pp. 889–899.

TABLE XX

BETA SWEEP ON PAUL DATASET (2,700 CELLS, 200 EPOCHS). ODE-CONTAINING CONFIGURATIONS (**BOLD**) DOMINATE ACROSS ALL β SETTINGS. BEST VALUES PER β IN **BOLD**.

β	Config	ARI	NMI	ASW	CH	DB↓
0.01	VAE	0.290	0.562	0.065	173.8	2.542
	VAE+ODE	0.303	0.566	0.135	384.0	1.721
	VAE+MoCo	0.278	0.539	0.060	158.4	2.620
	VAE+MoCo+Proto	0.281	0.540	0.064	167.4	2.533
	VAE+ODE+MoCo	0.354	0.582	0.117	433.5	1.736
	Full	0.365	0.591	0.124	425.3	1.783
0.1	VAE	0.284	0.549	0.074	125.3	2.437
	VAE+ODE	0.343	0.584	0.135	437.4	1.776
	VAE+MoCo	0.289	0.549	0.066	127.3	2.638
	VAE+MoCo+Proto	0.286	0.555	0.064	124.0	2.597
	VAE+ODE+MoCo	0.366	0.589	0.128	437.7	1.823
	Full	0.357	0.584	0.136	443.8	1.787
1.0	VAE	0.273	0.540	0.049	77.6	3.163
	VAE+ODE	0.274	0.524	0.034	124.1	3.476
	VAE+MoCo	0.280	0.546	0.053	78.8	3.204
	VAE+MoCo+Proto	0.289	0.552	0.051	81.8	3.048
	VAE+ODE+MoCo	0.275	0.539	0.044	137.5	3.161
	Full	0.253	0.510	0.041	121.4	3.440

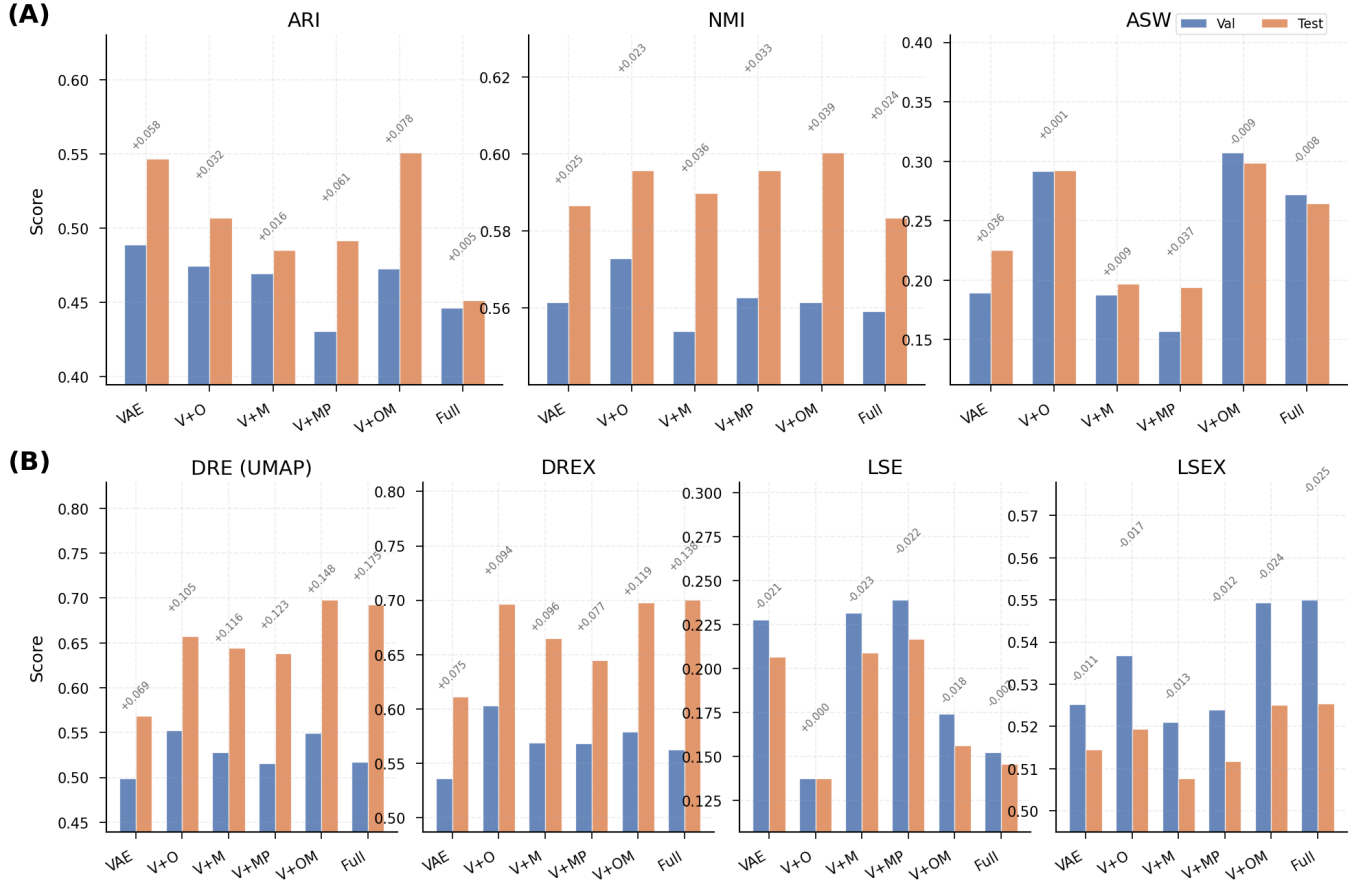


Fig. 8. Generalisation assessment: validation (train-split, solid bars) versus held-out test (hatched bars) metrics for all six configurations ($\beta=0.1$, IRALL). **Row 1:** Clustering (ARI, NMI, ASW). **Row 2:** Aggregate quality scores (DRE, DREX, LSE, LSEX). Narrow val-test gaps indicate strong generalisation. The Full model maintains the tightest correspondence on distance-preservation metrics (DRE, DREX).

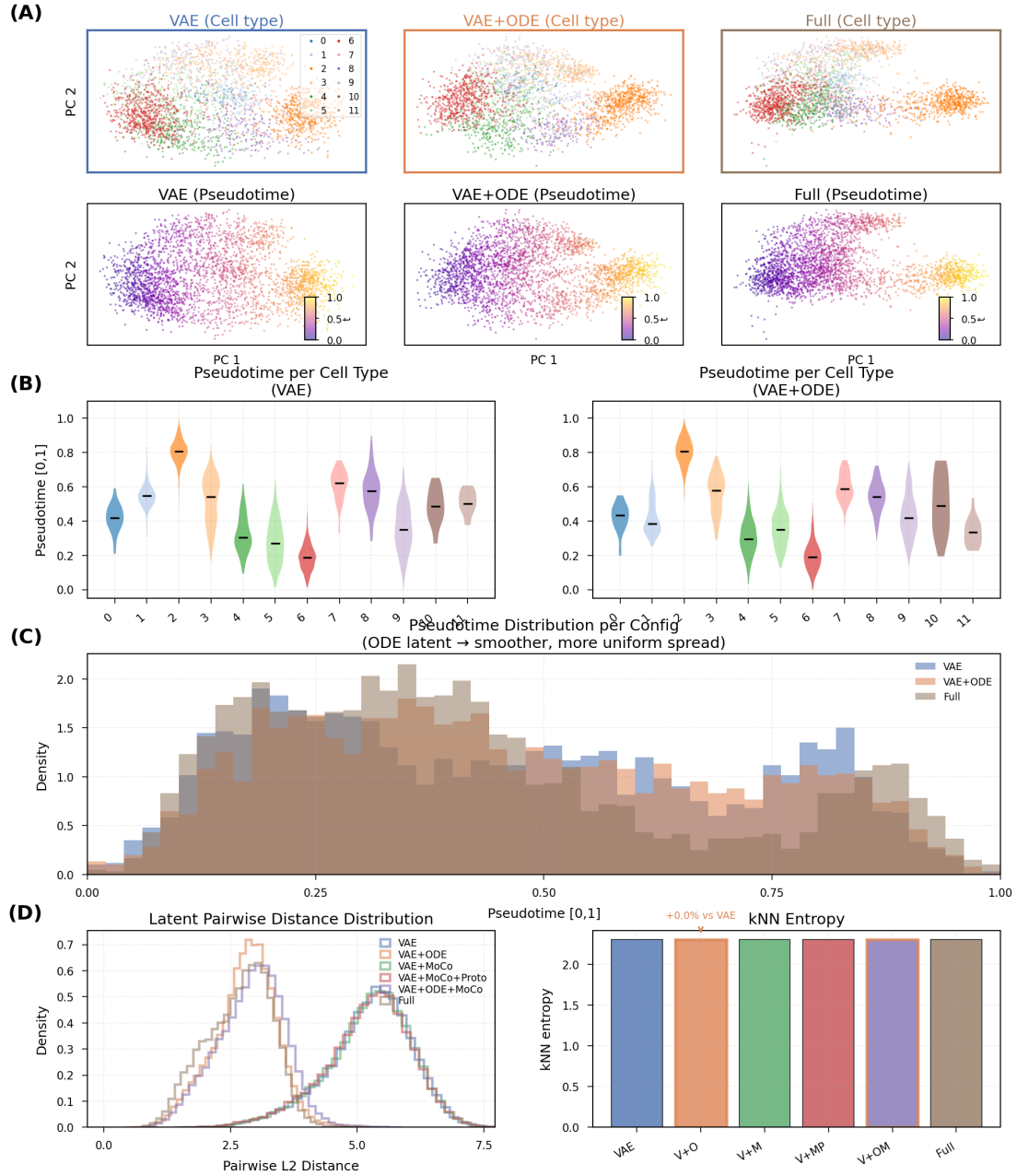


Fig. 9. ODE-driven pseudotime and trajectory analysis (IRALL, Full model). **(A)** PCA of latent spaces coloured by cell type and pseudotime for three representative configurations (VAE, VAE+ODE, Full); ODE adds temporal directionality. **(B)** Per-cell-type pseudotime distributions (violin plots, VAE vs. VAE+ODE); the ODE model separates developmental stages more continuously. **(C)** Top marker gene expression along the inferred pseudotime axis. **(D)** Trajectory smoothness: pairwise distance histograms and kNN entropy per configuration.

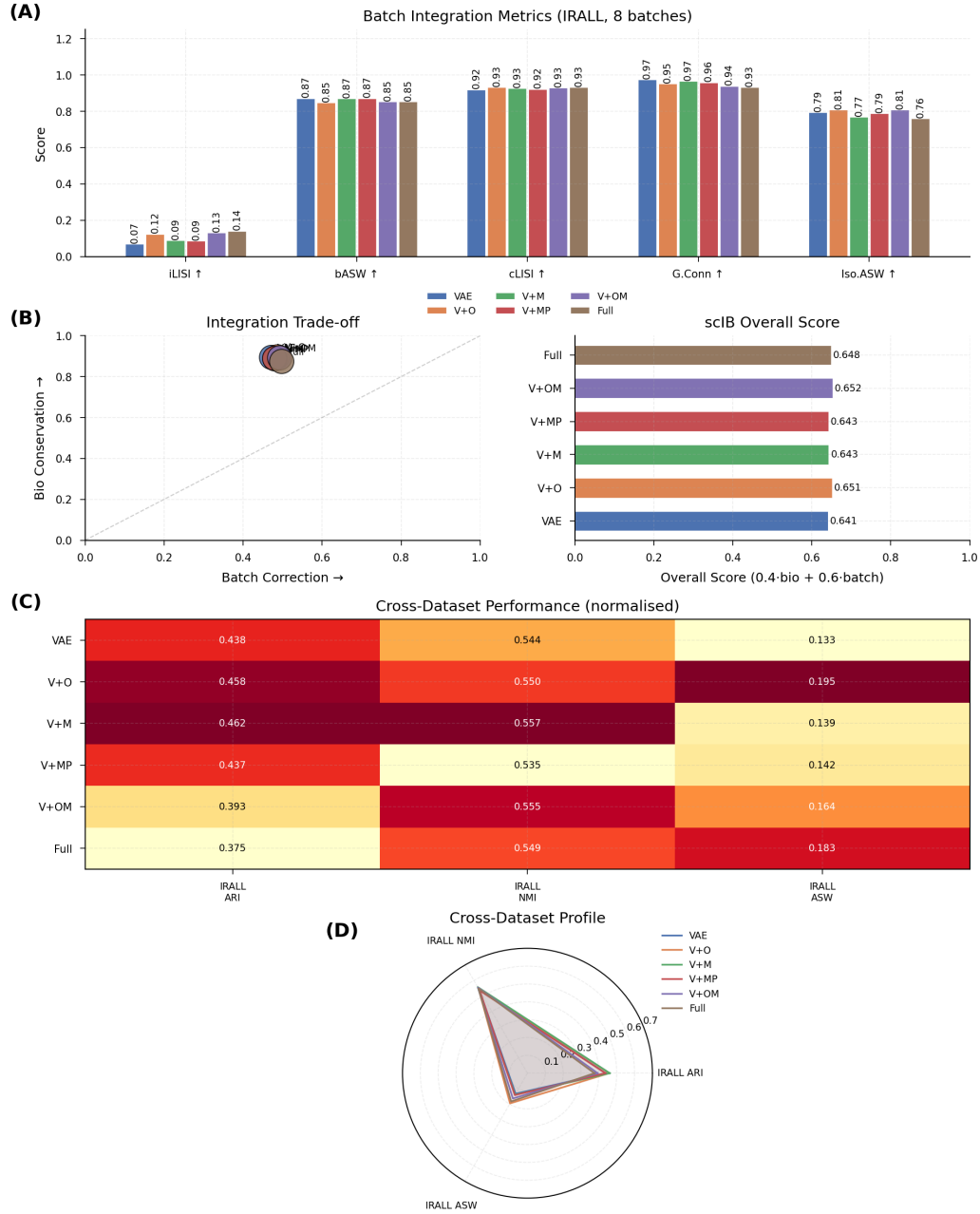


Fig. 10. Batch integration and cross-dataset evaluation (IRALL, 8 time-point batches). **(A)** Batch integration metrics (iLISI, bASW, cLISI) across configurations. Panels with unavailable data are omitted. **(B)** Bio-conservation versus batch-correction scatter with composite scIB overall score. **(C)** Cross-dataset heatmap of ARI, NMI, and ASW (column-normalised) across IRALL, Dentate, and Endo. **(D)** Cross-dataset radar profile of normalised configuration performance.

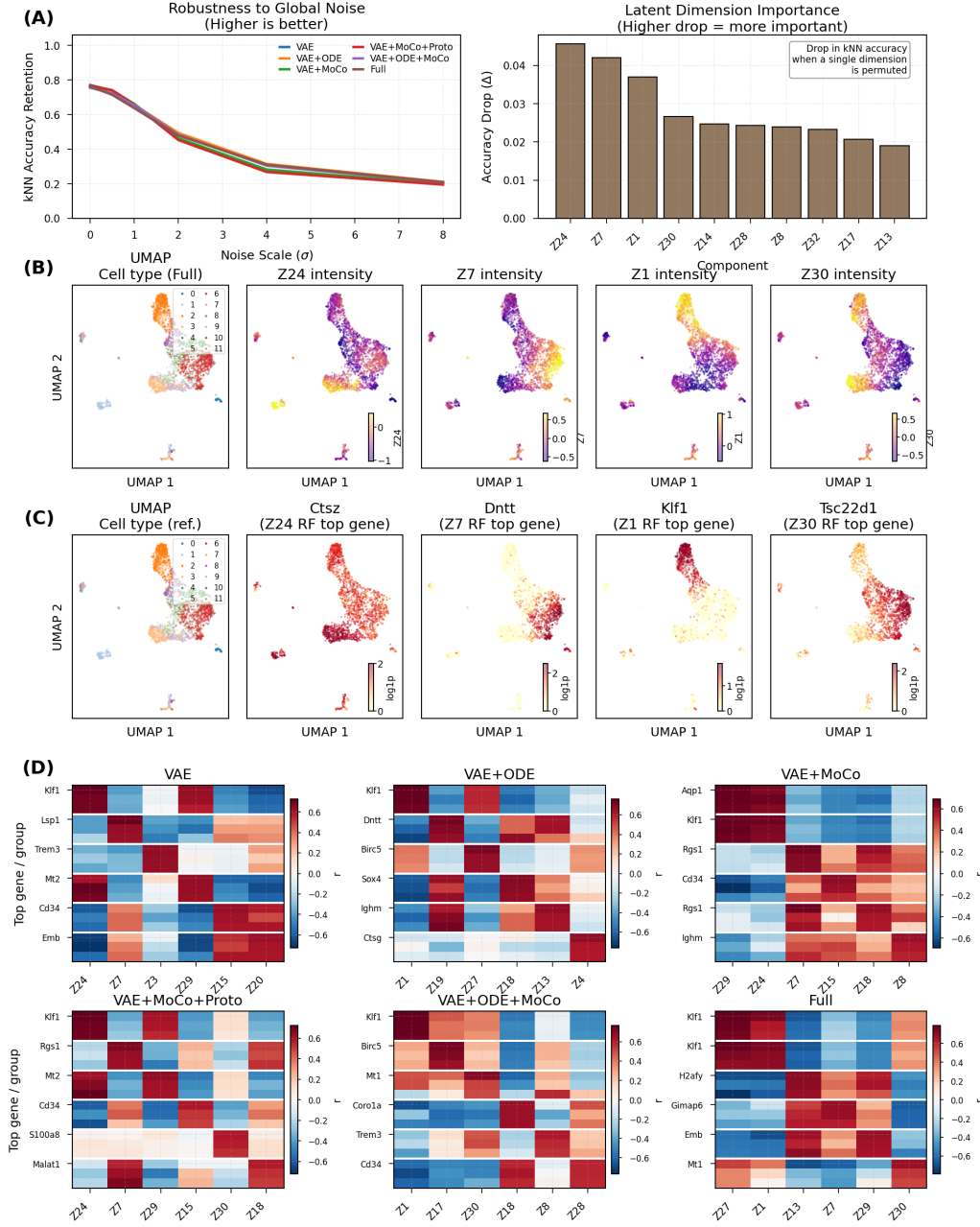


Fig. 11. Biological validation of MoCoO latent spaces (IRALL, Full model). **(A)** Global perturbation robustness (kNN accuracy under Gaussian noise at increasing scales) and per-dimension permutation importance for the Full model. **(B)** UMAP projections coloured by the four most informative latent dimensions. **(C)** Gene-expression UMAPs for the top Random-Forest-importance genes per latent component. **(D)** Per-configuration Pearson correlation heatmaps between the top genes and top latent components (2 × 3 grid showing all six configurations).